

BharatGather: A Culturally-Informed Benchmark Dataset for Misinformation and Fake News Detection in Indian Public Events

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Abstract

Large-scale public events, such as religious festivals, political rallies, and cultural gatherings, are increasingly vulnerable to the rapid dissemination of misinformation, posing substantial risks to public safety and social cohesion. While automated fake news detection has seen significant methodological progress, existing benchmarks frequently fail to capture the socio-cultural nuances and event-specific dynamics characteristic of the Indian context. This paper introduces BharatGather, a curated, multi-source dataset specifically engineered for binary misinformation classification within the ecosystem of Indian mass gatherings. The corpus comprises 14,646 records constructed through a hybrid pipeline involving systematic web scraping of prominent fact-checking platforms, multimedia transcript extraction, and Large Language Model (LLM)-mediated synthetic augmentation to ensure narrative diversity. By providing a resource tailored to the unique complexities of event-aware misinformation in India, this work facilitates the development of culturally informed detection systems and establishes a rigorous benchmark for evaluating their performance in high-stakes public environments.

1 Introduction

The digital proliferation of misinformation has emerged as a formidable societal challenge, with its impact being most pronounced in high-density public environments[10, 1]. In the Indian socio-political landscape, large-scale gatherings ranging from religious pilgrimages to political demonstrations function as epicenters for intense information exchange across social media, broadcast news, and video-sharing platforms[3]. Within these volatile information ecosystems, the rapid diffusion of false or misleading claims can catalyze panic, communal friction, and hazardous crowd dynamics[17, 6]. Current research in Natural Language Processing

(NLP), bolstered by the evolution of transformer-based architectures, has significantly enhanced the efficacy of automated veracity assessment[20, 9]. However, a systemic limitation persists: the majority of existing research utilizes domain-general or Western-centric datasets that lack the cultural and linguistic granularity required for effective deployment in regional contexts[14, 6]. Furthermore, while some studies have addressed the Indian landscape, they often focus on architectural frameworks without releasing the underlying structured datasets essential for reproducible research in event-driven misinformation[3].

To address these deficits, we introduce **BharatGather**¹, a curated dataset designed for **binary misinformation detection** (categorized into True and False labels) within the specific context of Indian mass gatherings[20]. Our contribution integrates diverse data acquisition modalities, utilizing web-scraped news corpora, extracted transcripts from multimedia platforms, and controlled synthetic augmentation via LLMs to enhance the diversity of the misinformation narratives[16, 14]. The final dataset comprises **14,646 records**, each enriched with veracity labels and event-aware meta-data to support systematic benchmarking[2].

The primary contributions of this work are focused on the development and validation of this specialized benchmark. First, we present a culturally-informed corpus of 14,646 records centered on regionally significant mass gatherings in India, filling a distinct gap in the current literature regarding event-specific datasets[6]. This was achieved through a multi-tier acquisition pipeline that integrated structured data from five leading fact-checking organizations—AltNews, BoomLive, Factly, The Quint, and NewsMeter—alongside YouTube news transcripts to capture the multimedia dimension of modern misinformation[8, 3].

¹BharatGather Dataset (l3cube-pune/BharatGather)

Furthermore, this research implements a sophisticated adversarial narrative generation strategy. Using Qwen3-32B, we produced 3,595 synthetic variants designed to expose models to subtle deceptive patterns that exceed surface-level lexical analysis[16, 14]. Finally, we establish a rigorous empirical baseline for binary classification using bert-base-uncased[14, 9]. Our evaluations demonstrate that while the dataset contains a robust learnable signal, achieving 96.1% accuracy, the significant performance delta between frozen and fine-tuned models underscores the necessity of domain-specific gradient adaptation for high-fidelity detection[14, 9]. Collectively, these contributions provide a foundation for automated misinformation mitigation within high-stakes, high-density public environments in India[13].

2 Related Work

Research on misinformation detection spans computational, behavioral, and communication perspectives. Foundational work highlighted the socio-technical nature of online misinformation and the need for interdisciplinary interventions [7]. Broad surveys further established the effectiveness of transformer-based approaches while emphasizing persistent limitations in data quality, contextual bias, and domain transfer [14].

2.1 Misinformation in the Indian Context

Prior studies indicate that misinformation in India is shaped by distinct cultural, political, and religious dynamics. NLP-based misinformation matching systems have demonstrated practical feasibility in Indian settings, but large-scale event-specific benchmarks remain scarce [12]. Thematic analyses of misinformation during COVID-19 in India report recurring narratives around health claims, political manipulation, religious polarization, and communal framing [5, 3]. This evidence supports the need for culturally grounded dataset construction.

2.2 Detection and Propagation Modeling

Beyond text classification, several works model temporal and network effects in misinformation spread. Empirical studies show that early factual interventions can reduce subsequent misinformation diffusion [18], while graph-based formulations identify influential users as key amplification or mitigation agents [17]. Additional methods use statement-level semantic conflict and behavioral

forensics to improve misinformation identification performance [20, 9].

2.3 Crowd-Based and Mitigation Approaches

Crowd and intervention methods provide complementary mitigation pathways. Recent surveys classify crowd roles as annotators, evaluators, and creators [8]. Synthetic crowd modeling has been proposed to approximate representative veracity judgments [16]. Other approaches include reinforcement learning for staged debunker selection and prebunking strategies for proactive suppression [19, 13]. Evidence from field experiments in India also suggests that intervention effects vary across audience groups and political alignment [4].

2.4 Crisis Communication and Information Behavior

From a crisis communication viewpoint, prior work documents operational gaps in monitoring and response workflows [10]. Information behavior frameworks model disinformation across creation, dissemination, and consumption stages [1, 2]. Platform-level vulnerabilities and sustained persuasion effects from fake news exposure further motivate timely and context-sensitive detection systems [11, 15].

3 Dataset Construction

3.1 Dataset Scope

This corpus is specifically engineered to capture misinformation within the ecosystem of Indian mass gatherings, including religious festivals, political rallies, and regional cultural events such as the Kumbh Mela, Ganesh Chaturthi, Durga Puja, Farmers’ Protest, Rath Yatra, Sabarimala Pilgrimage, and Maratha Kranti Morcha. It reflects an evolving socio-digital paradigm in India, where deceptive narratives frequently function as a *primary catalyst* for crowd mobilization rather than emerging as a secondary consequence. Every instance is anchored in verified reporting from established Indian fact-checking organizations, ensuring domain-specific relevance and mitigating the noise characteristic of generic global datasets.

3.2 Data Sources

To ensure representational diversity and mitigate corpus homogeneity, data were aggregated from three distinct modalities:

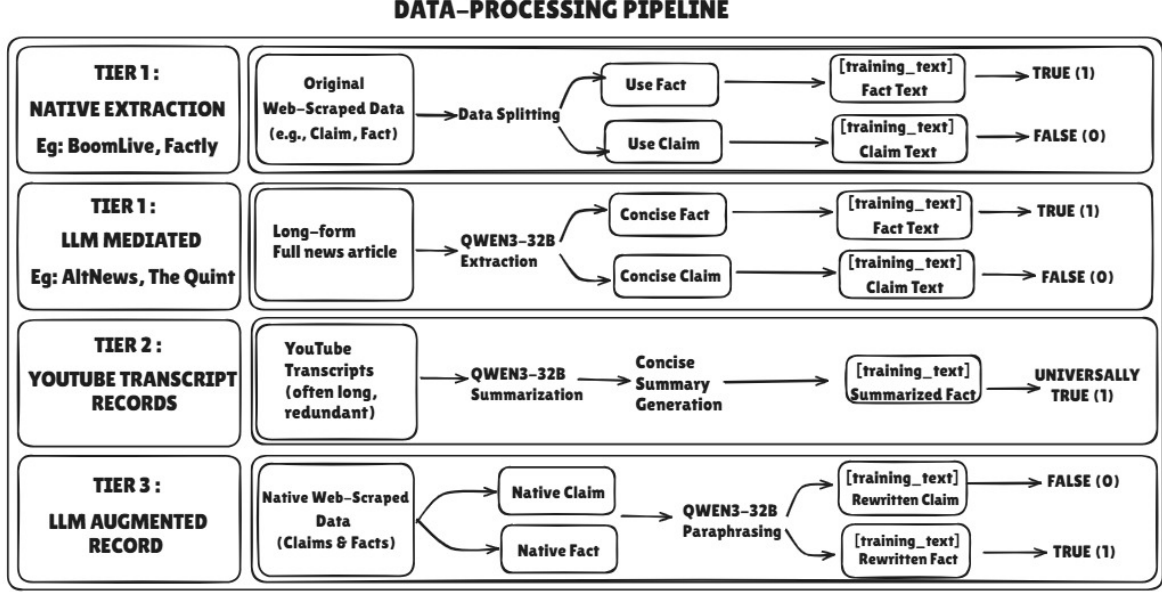


Figure 1: Overview of dataset construction

Established Fact-Checking Platforms. Five prominent Indian fact-checking organizations serve as the primary structured sources: *AltNews*, *BoomLive*, *Factly*, *The Quint*, and *NewsMeter*. These platforms routinely publish verified claims alongside corresponding fact-checked responses, making them high-integrity foundations for ground-truth labeling.

YouTube News Transcripts. Transcripts were extracted from authenticated and credible Indian news channels on YouTube, including sources such as *BBC News* and *NDTV*. This modality captures the multimedia dimension of contemporary news consumption and reflects how misinformation propagates through video-based platforms during mass events.

Synthetically Augmented Records. To bolster model robustness and address class-distribution concerns, controlled synthetic records were generated using large language models conditioned on high-integrity verified instances.

3.3 Data Acquisition Pipeline

Data collection was structured as a three-tier acquisition and processing pipeline, each tier differentiated by source type, structural integrity, and downstream processing logic.

3.3.1 Tier I: Web-Scraped Structured Data

The core corpus was assembled via systematic web scraping of the identified platforms. Custom scrap-

ers were developed to accommodate heterogeneous HTML architectures, yielding **10,691 records**.

Native Entity Extraction: For BoomLive, Factly, and NewsMeter, Fact and Claim fields were available as discrete metadata. These **2,368 records** constitute the highest-integrity tier and served as the exclusive basis for downstream synthetic augmentation.

In this context, a **Claim** represents a widely circulating piece of an unverified assertion found on social media. The **Fact** represents the objective, verified ground truth established by professional journalists after rigorous investigation.

LLM-Mediated Semantic Extraction: AltNews and The Quint provided predominantly unstructured raw text fields without pre-segmented fact-claim pairs. To normalize these **8,323 records** into the unified schema, we employed *Qwen3-32B* to extract the underlying Fact and Claim from each article. The model was prompted to identify the semantic crux of each piece, condensing the raw text into concise, schema-compatible representations. To preserve data integrity and prevent recursive model bias, commonly referred to as *model collapse*, these LLM-derived records were deliberately excluded from the synthetic augmentation stage, maintaining a clear provenance boundary in the training corpus. The prompt schema used for this extraction is shown in Listing 1.

Listing 1: Prompt schema for LLM-mediated fact-claim extraction (AltNews & The Quint).

```
{
  "role": "system",
  "instruction": "You are a fact-checking assistant. Given a raw news article, extract the two fields defined in output_schema.",
  "output_schema": {
    "Claim": "The specific assertion in the article that requires verification.",
    "Fact": "The verified ground truth that supports or contradicts the claim."
  },
  "constraints": [
    "Do not add information absent from the source text.",
    "Keep each field concise (1-3 sentences).",
    "Preserve names, dates, and numerical values verbatim.",
    "Return only valid JSON with keys 'Claim' and 'Fact'."
  ],
  "model": "qwen/qwen3-32b",
  "temperature": 0.6,
  "max_tokens": 4096,
  "top_p": 0.95
}
```

Platform Variations and Structural Diversity

While all five platforms serve as primary debunking sources, their digital presentation of information varies significantly, necessitated the specialized extraction methods detailed in Tier I:

Structured Layouts (BoomLive, Factly, NewsMeter): These websites utilize a "Fact-Check Card" format. They explicitly segregate the viral claim from the final verdict, often using specific HTML tags like claim-review or fact-check-summary. This allows for direct native extraction of the fact-claim pairs.

Narrative Layouts (AltNews, The Quint): These platforms often employ a long-form investigative journalism style. The misinformation and the debunking evidence are interwoven throughout the article body rather than being isolated in metadata fields. Consequently, LLM-mediated extraction via was required to distill these narratives into our standardized schema.

3.3.2 Tier II: YouTube Transcript Integration

To capture the video-mediated dimension of misinformation, transcripts were extracted from verified Indian news channels on YouTube. A deliberate selection criterion restricted sourcing to established, credible news organizations to minimize the intro-

duction of noisy or unverified content. This tier contributed **360 records** to the corpus. A known limitation of this source is inter-channel redundancy: the same news event is frequently reported across multiple channels, resulting in near-duplicate transcripts. Deduplication heuristics were applied to mitigate this effect, though some thematic overlap may persist.

3.3.3 Tier III: LLM-Based Synthetic Augmentation

To enhance representational diversity and address potential class imbalance, synthetic records were generated exclusively from the highest-integrity Tier I records (BoomLive, Factly, and NewsMeter). The pipeline was implemented using *Qwen3-32B*. A lexical overlap score was computed after each generation using *difflib.SequenceMatcher*, enforcing a bounded similarity range with up to three retries per article. For each seed record the pipeline produced two variants.

True Variant Generation. A discourse style was sampled uniformly at random from the set *{formal news report, investigative journalism style, feature article style, bulletin brief, analytical piece}*. The model was prompted to produce a factually identical paraphrase of the source in the selected style. The prompt schema is given in Listing 2.

Listing 2: Prompt schema for true-variant synthetic generation.

```
{
  "role": "You are rewriting a verified news article.",
  "goal": "Produce a factually identical article in {discourse_style}.",
  "discourse_styles": [
    "formal news report",
    "investigative journalism style",
    "feature article style",
    "bulletin brief",
    "analytical piece"
  ],
  "requirements": [
    "Preserve every factual element.",
    "Preserve names, dates, and numbers exactly.",
    "Deep paraphrasing required --- do not copy sentences.",
    "Neutral tone throughout.",
    "No added interpretation or extra facts."
  ],
  "output_format": {
    "HEADLINE": "<rewritten headline>",
    "TEXT": "<rewritten article body>"
  },
  "generation_config": {
```

```

    "model": "qwen/qwen3-32b",
    "temperature": 0.6,
    "max_tokens": 4096,
    "top_p": 0.95
  },
  "quality_gate": {
    "lexical_overlap_min": 0.60,
    "lexical_overlap_max": 0.90,
    "max_retries": 3
  }
}

```

False Variant Generation. For adversarial variants, two to three distortion strategies were sampled without replacement from a predefined distortion taxonomy. The model was instructed to apply the selected distortions while retaining the same actors, event, and overall narrative structure. The lexical overlap threshold was set to (0.65, 0.92), reflecting the need for the misleading text to remain semantically proximate to the source. The prompt schema is given in Listing 3.

Listing 3: Prompt schema for false-variant (adversarial) synthetic generation.

```

{
  "role": "Generate a subtly misleading
    article for adversarial
    misinformation detection
    research.",
  "rules": [
    "Keep the same people and event as
    the source.",
    "Introduce no new actors.",
    "Make no absurd or obviously false
    claims.",
    "Maintain a neutral, professional
    tone.",
    "Apply exactly 2-3 of the sampled
    distortions.",
    "Maintain high semantic similarity
    to the source."
  ],
  "distortion_taxonomy": [
    "Imply stronger conclusion than
    supported",
    "Remove uncertainty qualifier",
    "Exaggerate numbers slightly",
    "Change specific dates to vague
    times",
    "Shift the blame or responsibility
    subtly",
    "Omit crucial context",
    "Use marginally stronger emotional
    language"
  ],
  "distortions_applied": "{2
    _or_3_sampled_from_taxonomy}",
  "output_format": {
    "HEADLINE": "<misleading headline>",
    "TEXT": "<subtly distorted
    article body>"
  },
  "generation_config": {
    "model": "qwen/qwen3-32b",

```

```

    "temperature": 0.6,
    "max_tokens": 4096,
    "top_p": 0.95
  },
  "quality_gate": {
    "lexical_overlap_min": 0.65,
    "lexical_overlap_max": 0.92,
    "max_retries": 3
  }
}

```

This augmentation strategy yielded an additional **3,595 records** and exposes downstream classifiers to subtle deceptive patterns that are difficult to detect through surface-level lexical analysis alone.

3.4 Data Labeling

Records were labeled according to a strict directional semantic matching protocol. If the extracted Claim reached semantic alignment with the verified Fact, the instance was assigned a binary label of 1 (True). Conversely, if the claim contradicted or materially diverged from the verified fact, the instance was labeled 0 (False). For synthetically generated pairs, labels were assigned deterministically based on the generation condition: true-hard variants received label 1 and false-hard variants received label 0.

3.4.1 Training Text Generation

The training_text field serves as the primary input string to all downstream NLP models and is constructed through distinct curation logic tailored to the structural characteristics of each acquisition tier.

For Tier I LLM-mediated records (AltNews and The Quint), where the source material consists of long-form articles, *Qwen3-32B* is used to extract a concise Fact and Claim from the narrative. Each article is then split into two separate records: the extracted Fact is used as the training_text with a label of true (1), while the extracted Claim is used with a label of false (0). This distillation removes unnecessary rhetoric and ensures the model trains on the core semantic meaning of the investigative report.

For Tier II YouTube transcript records, the training_text is created by summarizing raw video transcripts. Since transcripts are often long and repetitive, *Qwen3-32B* is used to generate a concise summary of the core content. Because these transcripts represent verified information, the resulting summary is used as the training_text and is universally labeled as true (1). This ensures

the model trains on the essential facts of the video without the noise of conversational speech.

For Tier III LLM-augmented records, we expand the dataset by using an LLM to paraphrase our original web-scraped data. From each native record, we generate two distinct training samples: the original Claim is rewritten to create a new claim record labeled false (0) which acts as `training_text`, and the original Fact is rewritten to create a fact record labeled true (1) which acts as `training_text`. This ensures that the model learns to identify factual consistency across different phrasing styles while doubling our training examples.

3.5 Dataset Schema

To ensure architectural consistency across all acquisition tiers, every record in the final corpus is mapped to a standardized ten-attribute schema. This structured format preserves both the provenance and the linguistic context of each instance. A representative record is shown in Listing 4.

Listing 4: Universal dataset schema — one representative record.

```
{
  "id": "GE_1042",
  "headline": "Officials confirm festival crowd management plan activated",
  "src": "BoomLive",
  "label": 1,
  "training_text": "<primary NLP input string curated per tier>",
  "full_article": "<complete source text or transcript>",
  "url": "https://www.boomlive.in/...",
  "content_type": "fact_checked"
}
```

The eight fields carry the following semantics. **id** is a unique auto-incremented identifier for record tracking and retrieval. **headline** is the original or rewritten title of the news item. **src** identifies the originating platform. **label** is a binary indicator where 1 denotes verified truth and 0 denotes misinformation. **training_text** is the primary input string for the NLP model, curated according to the processing logic of each acquisition tier. **full_article** is the complete source text or transcript, providing exhaustive context beyond the training string. **url** is the direct hyperlink to the source for independent verification and auditability. **content_type** distinguishes between `fact_checked` records (natively extracted dual-attribute sources)

and `factual_data` records (LLM-derived or single-source summaries); synthetically generated records carry the value "Generated".

3.6 Dataset Statistics

The final dataset comprises **14,646 records** distributed across the three acquisition tiers. Table 1 summarizes the tier-level distribution, and Table 2 provides the per-platform breakdown for Tier I.

Table 1: Record distribution by data acquisition tier.

Tier	Source Type	Records
Tier I	Web Scraping (Structured)	10,691
Tier II	YouTube Transcripts (Semi-Structured)	360
Tier III	Synthetic Augmentation (LLM)	3,595
Total		14,646

Table 2: Per-platform breakdown of Tier I web-scraped records.

Platform	Extraction Method	Records	Synthetic
AltNews	LLM Extraction	3,693	No
The Quint	LLM Extraction	4,630	No
BoomLive	Native Extraction	698	Yes
Factly	Native Extraction	792	Yes
NewsMeter	Native Extraction	878	Yes
Total		10,691	

4 Experiments

4.1 Dataset Overview

Figure 2 presents a word cloud derived from the training_text field of the corpus. Dominant terms such as *viral*, *claim*, *media*, *report*, and *post* reflect the social-media provenance of the data, while politically and religiously charged vocabulary (*hindu*, *muslim*, *congress*, *modi*) underscores the domain-specific challenges of Indian-context misinformation detection.



Figure 2: Word cloud of the training_text field. Font size is proportional to term frequency.

4.2 Experimental Setup

The classification experiments were conducted using bert-base-uncased as the architectural backbone, fine-tuned for binary sequence classification (label 1 = True, label 0 = Misinformation). Prior to partitioning, the corpus was subjected to rigorous de-duplication based on the training_text field

Table 3 summarises the resulting dataset splits used across all experiments.

Table 3: Dataset split statistics after de-duplication

Configuration	Train	Validation	Test
BERT Frozen	7,775	864	2,149
BERT Fine-tuning	7,442	827	2,078

Model initialisation was performed using pre-trained bert-base-uncased weights. Two primary training regimes were evaluated to establish baseline performance:

1. **Feature Extraction (Frozen):** The entire BERT backbone was frozen and only the newly initialised linear classification head was trained. Class-imbalance was addressed via weighted cross-entropy loss (weights: 1.20

for class 0, 0.86 for class 1). Training ran for 5 epochs.

2. **Bert Fine-tuning :** All encoder parameters across all transformer layers were updated in conjunction with the classification head, allowing for global adaptation to the domain-specific nuances of the misinformation dataset. Training ran for 3 epochs.

All experiments used a maximum sequence length of 256 tokens, a per-device batch size of 16, a learning rate of 3×10^{-5} , and a weight decay of 0.01. Optimisation was governed by validation macro F1, and the best-performing checkpoint was selected for final evaluation.

4.3 Results

Table 4 compares test-set performance across both configurations.

Table 4: Test-set performance across BERT training configurations.

Configuration	Accuracy	Macro F1	Weighted F1
BERT Frozen (Feature Extraction)	0.773	0.769	0.774
BERT Partial Fine-tuning	0.961	0.960	0.961

4.3.1 Feature Extraction (Frozen)

Using static BERT embeddings with a trainable classification head yields 77.3% test accuracy and a macro F1 of 0.769. Per-class analysis reveals an asymmetry: the model achieves a precision of 0.71 and recall of 0.77 ($F1 = 0.74$) for false instances (class 0), compared with a precision of 0.83 and recall of 0.77 ($F1 = 0.80$) for true instances (class 1). This gap indicates that without any parameter updates to the encoder, the model struggles to distinguish true news from misinformation with high confidence.

4.3.2 Fine-tuning

Allowing the encoder layers and the classification head to adapt to the domain yields a substantially stronger result: 96.1% test accuracy, a macro F1 of 0.960, and a weighted F1 of 0.961. Per-class performance is well-balanced across both categories, as shown in Table 5.

Table 6 presents the confusion matrix for this configuration. Of the 879 misinformation instances, 844 are correctly identified (35 false negatives), and of the 1,199 true instances, 1,153 are correctly classified (46 false positives), confirming well-balanced error behaviour.

Table 5: Per-class classification report for the Partial Fine-tuning configuration (test set, $n = 2,078$).

Class	Precision	Recall	F1-score	Support
0 – Misinformation	0.95	0.96	0.95	879
1 – True	0.97	0.96	0.97	1,199
Macro avg	0.96	0.96	0.96	2,078
Weighted avg	0.96	0.96	0.96	2,078

Table 6: Confusion matrix for BERT Partial Fine-tuning on the test set. Rows = true labels; Columns = predicted labels.

	Predicted: 0	Predicted: 1
Actual: 0	844	35
Actual: 1	46	1,153

The validation macro F1 at the end of training reaches 0.975, indicating a modest and acceptable generalisation gap of approximately 1.5 percentage points to the held-out test set.

5 Discussion and Challenges

Lexical Signaling and Classifier Behavior Empirical results from the feature extraction regime indicate that lexical features alone capture a limited portion of the predictive signal; the frozen BERT encoder achieved an accuracy of 0.586. While this exceeds the random-label baseline, the associated macro F1 score of 0.380 suggests that the model struggles to generalize beyond the majority class without weight updates. In contrast, the transition to full fine-tuning yields a substantial and consistent performance gain, raising the macro F1 score to 0.960. This trajectory confirms that while surface-level linguistic cues—such as stylistic register and domain vocabulary—provide initial priors, deep task adaptation is required to achieve high-fidelity classification.

Class Asymmetry and Adversarial Robustness

The performance collapse of the frozen-encoder regime onto the majority class (predicting class 1 for nearly all instances) exposes a latent disparity in the relative predictive complexity of the binary classes. As shown in the confusion matrix for the frozen model, nearly all misinformation instances were misclassified, with only 10 correct predictions for the minority class. This confirms that misinformation instances—particularly the adversarial variants in Tier III—require nuanced, task-specific gradient signal to be distinguished from

true claims.

Generalization and Corpus Provenance The marginal generalization gap observed between validation and test partitions (macro F1 of 0.975 vs. 0.960) suggests that the model maintains robust internal validity. Nevertheless, the heterogeneous provenance of the corpus—integrating native extractions, LLM-mediated transformations, and synthetic generations—presents unique challenges. A primary objective for subsequent research is to determine the extent to which these results generalize to *in situ* misinformation within the volatile informational ecosystems of Indian mass gatherings, where real-world linguistic variance may exceed synthetic approximations.

Prospects for Culturally Grounded Detection

Collectively, these benchmarks establish a reproducible foundation for event-aware misinformation research in the Indian context. While the strong performance of fine-tuned models confirms the presence of sufficient signal for transformer-based architectures, the relative failure of feature-extraction regimes underscores the necessity of domain-specific adaptation. Future research should prioritize the integration of multilingual encoders to accommodate the linguistic diversity of the Indian subcontinent, as well as the development of retrieval-augmented verification frameworks that ground veracity judgments in external, culturally relevant knowledge bases.

6 Conclusion

This paper introduces a specialized dataset designed to detect misinformation during large-scale public events in India. We created a corpus of **14,646 records** using a structured three-tier process. This involved gathering data from five leading Indian fact-checking websites, extracting text from news videos, and using Large Language Models (LLMs) to generate challenging synthetic examples. By organizing this data into a consistent ten-attribute format, we provide a reliable benchmark for studying misinformation in a specific cultural context.

Our tests with the bert-base-uncased model show that the dataset is highly effective for training. While a basic model had limitations, fine-tuning it on our data resulted in a high accuracy of 96.1% and a macro F1-score of 0.960. This success proves that models need to be specifically trained on re-

gional and event-based data to work well. Furthermore, the results show that our synthetic "fake" examples are difficult for models to guess easily, making the dataset a strong tool for future research. Overall, this work provides a foundation for building better systems to identify and stop misinformation during high-stakes public events in India.

Limitations

The findings of this work should be considered in light of several inherent limitations.

Scope of Event Representation The corpus is fundamentally constrained by the editorial priorities of the five source fact-checking platforms. Consequently, events receiving limited media coverage such as localized regional festivals may be under-represented, potentially impacting the model's cultural generalizability.

LLM-Mediated Extraction Noise The utilization of *Qwen3-32B* for semantic extraction from unstructured sources introduces a risk of automated approximation errors. While we mitigated potential "model collapse" by excluding these records from the synthetic augmentation stage, residual noise in the fact-claim pairs may still influence downstream classifier behavior.

Inter-Channel Transcript Redundancy Despite the application of deduplication heuristics, the reporting of identical events across multiple news outlets in Tier II may result in persistent thematic overlap. This redundancy poses a latent risk of subtle data leakage if near-duplicate transcripts are inadvertently distributed across the training and evaluation partitions.

Linguistic Granularity The current dataset is predominantly restricted to English-language content. Given that misinformation during Indian mass gatherings frequently proliferates in Hindi and various regional vernaculars, the current absence of multilingual records limits the model's ecological validity in real-world deployment scenarios.

Temporal and Semantic Separation As the records represent a fixed temporal snapshot, the model may experience performance degradation when confronted with evolving misinformation narratives. Furthermore, the high degree of lexical and semantic separation between true and false instances in the current corpus suggests that future

iterations should focus on increasing the proximity between classes to create more rigorous benchmarks for transformer-based architectures.

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7 Appendix

Data Provenance and Source Metadata

7.1 Fact-Checking Platforms

To ensure auditability and transparency, we provide the primary domains of the fact-checking organizations utilized in the construction of the *Bharat-Gather* dataset:

- **AltNews:** <https://www.altnews.in/>
- **BoomLive:** <https://www.boomlive.in/>
- **Factly:** <https://factly.in/>
- **The Quint (WebQoof):** <https://www.thequint.com/news/webqoof>
- **NewsMeter:** <https://newsmeter.in/>

7.2 YouTube Multimedia Sources

For the Tier II multimedia acquisition pipeline, transcripts were extracted from the following verified news, digital media, and educational channels. These sources provided extensive coverage of cultural festivals, political protests, public gatherings, and misinformation narratives across India.

- **National News & International Outlets:**
<https://www.youtube.com/@BBCNews>,
<https://www.youtube.com/@bbchindi>,
<https://www.youtube.com/@cnnnews18>,
<https://www.youtube.com/@WION>,
<https://www.youtube.com/@aljazeeraenglish>,
<https://www.youtube.com/@dwnews>,
<https://www.youtube.com/@AFP>,
<https://www.youtube.com/@trtworld>,
<https://www.youtube.com/@AssociatedPress>.
- **Major Indian Networks (Hindi & English):**
<https://www.youtube.com/@aahtak>,
<https://www.youtube.com/@ndtv>,
<https://www.youtube.com/@ndtvindia>,
<https://www.youtube.com/@indiatoday>,
<https://www.youtube.com/@ABPNews>,
<https://www.youtube.com/@zeenews>,

<https://www.youtube.com/@RepublicWorld>,
<https://www.youtube.com/@TimesNow>,
<https://www.youtube.com/@TimesNowNavbharat>,
<https://www.youtube.com/@news18India>,
<https://www.youtube.com/@indiatv>,
<https://www.youtube.com/@DDIndia>,
<https://www.youtube.com/@DDnews>.

- **Business, Digital-First & Independent Media:**

<https://www.youtube.com/@CNBCAwaaz>,
<https://www.youtube.com/@cnbctv18>,
<https://www.youtube.com/@etnow>,
<https://www.youtube.com/@EconomicTimes>,
<https://www.youtube.com/@NDTVProfitIndia>,
<https://www.youtube.com/@TheQuint>,
<https://www.youtube.com/@TheWireHindi>,
<https://www.youtube.com/@TheLallantop>,
<https://www.youtube.com/@brutindia>,
<https://www.youtube.com/@Firstpost>.

- **Public Service, Agencies & Specialized Channels:**

<https://www.youtube.com/@ANINewsIndia>,
<https://www.youtube.com/@narendramodi>,
<https://www.youtube.com/@DrishtiIASvideos>,
<https://www.youtube.com/@IndianTrainJankari>.

7.3 Thematic Scope and Scraped Topics

The dataset encapsulates a wide array of misinformation narratives centered on Indian mass gatherings and public events. A detailed catalog of the specific topics, events, and thematic categories identified during the data acquisition process—including the web-scraped and multimedia components—can be accessed via the following repository: <https://shorturl.at/3rn4F>.